HO CHI MINH CITY UNIVERSITY OF ECONOMICS AND FINANCE (UEF)
FACULTY OF INFORMATION TECHNOLOGY

Real-Time Weather Data Collection & Cleaning

End-to-End Multi-API Harvesting, 9-Step Preprocessing & Empirical Quality Evaluation Across 20 Vietnam Cities

4 APIs / 5 Feeds	20 Locations	49,109 Rows	9 Clean Steps	44,540 Rows
Multi-Source Streams	7 Vietnam Climate Zones	Raw Input Records	Automated Preprocessing	100% Certified Output

CAPSTONE PROJECT METADATA, SUPERVISION & RESEARCH TEAM

Dimension	Details / Identification	Role & Responsibility in Capstone
Course & Class	Getting and Cleaning Data — 261.DAS1204E.A01E	Final Project Defense — Topic 6 (Real-Time Weather)
Lecturer	**MSc. Truong Ba Vinh (vinhtb@uef.edu.vn)**	Academic Supervisor — Faculty of IT (UEF)
Student 1	**Nguyen Da Giao Dang - 235210195**	Part 1: Multi-API Collection & Raw Dataset Audit
Student 2	**Ho Van Nhu Y - 245191885**	Part 2: 9-Step Data Cleaning & Sensor Fusion Pipeline
Student 3	**Ngo Minh Triet - 245193038**	Part 3: Data Quality Evaluation, Statistics & Dashboard
Dataset Scope	July 17, 2026 – October 17, 2026 (2,227 hours/city)	49,109 raw rows -> 44,540 clean rows (0% missing/dups)

Presentation Roadmap

Organized into 3 core parts — visual-first presentation with tables, charts & live evidence

PART 1

Data Collection & Dataset Description

4 APIs | 20 Cities | 49,109 Raw Rows

PART 2

Data Cleaning & Preprocessing

9 Steps | -4,569 Rows | 0 Defects Left

PART 3

Results & Data Quality Evaluation

44,540 Clean Rows | 3 Core Insights

STRUCTURED PRESENTATION MATRIX ACROSS THE 3 REQUIRED PARTS

Section	Core Questions Addressed	Key Visuals / Tables Presented	Quantitative Outcome
1.1 – 1.2	Where data comes from & how it is collected	Provider Table, 7-Zone Table & Collector Code	100 API calls -> 49,109 rows
1.3 – 1.4	What structure data has & raw data problems	22->31 Schema Table & 9-Defect Audit Table	11 missing calls, 3,574 dups
2.0 – 2.2	Missing responses, empty rows & UTC+7 time	9-Step Table, Coverage Heatmap & TS Code	995 future forecast rows cut
2.3 – 2.5	Physical ranges, 3-tier dedup, fusion & grid	WMO Bounds Table, Dedup Code & Grid Code	125 gust & 158k float fixed
3.1 – 3.3	Before-vs-after proof & output distributions	Before/After Chart, 9-Stat Table & Boxplots	44,540 rows (100% complete)
3.4 – 3.8	Correlations, 20-city climate & provider bias	Correlation/Rain, 20-City, Bias & Streamlit UI	r = -0.66; Xweather +1.49°C

Where Data Comes From

4 external weather providers (5 endpoint groups) across 20 locations in 7 Vietnam climate zones

4 APIs / 5 Endpoints

Multi-Source Weather Feeds

Combines ERA5 archive, NWP hourly forecast, and live station observations.

20 Vietnam Cities

3.3× Above Minimum Spec

Spans 7 climate zones from alpine Sa Pa (1,545m) to offshore Con Dao.

49,109 Raw Rows

93-Day Hourly Window

July 17 – Oct 17, 2026 (2,227 hours/city across 100 endpoint calls).

TABLE 1: WEATHER API PROVIDERS & ENDPOINT SHARE (N = 49,109)

Provider	Endpoint Group	Record Type	Succeeded Calls	Raw Rows	Share
Open-Meteo	archive (ERA5)	historical_hourly	20 / 20	**44,640**	90.90%
Open-Meteo	forecast	forecast_hourly	20 / 20	2,880	5.86%
WeatherAPI.com	forecast + current	hourly + current	15 / 20	1,095	2.23%
Vaisala Xweather	observations	recent_hourly	20 / 20	480	0.98%
FreeOpenAPI.dev	weather	current_snapshot	14 / 20	14	0.03%

TABLE 2: 20 MONITORING LOCATIONS ACROSS 7 VIETNAM REGIONS

Climatic Region	Monitored Cities / Stations	Elevation	Climate Regime
1. North Mountain	Sa Pa, Dien Bien Phu, Ha Long	7–1,545m	Highland & coastal bay
2. Red River Delta	Hanoi, Nam Dinh	3–20m	Monsoon delta plain
3. North Central	Vinh, Hue City, Phu Loc – Hue	8–1,002m	Foehn & heavy autumn rain
4. South Central	Da Nang, Nha Trang, Phan Thiet	9–11m	Warm dry coastal belt
5. Central Highland	Pleiku, Buon Ma Thuot, Da Lat	462–761m	Temperate plateau
6. South & Mekong	TP.HCM, Vung Tau, Can Tho, Ca Mau	6–13m	Sub-equatorial monsoon
7. Offshore Islands	Phu Quoc, Con Dao	92–180m	Oceanic buffered regime

How Data Is Collected

Automated Python multi-API ingestion workflow with retry resilience & coverage logging

Step 1: Config & Auth

20 Cities × 5 Endpoints

Loads coordinates, time windows & API keys from config/config.py.

Step 2: HTTP Retry

Exponential Backoff

3 retries on HTTP 429/5xx timeouts; catches provider dropouts safely.

Step 3: Dual Timestamps

collected_at vs. date

Tags exact retrieval time (collected_at) alongside valid weather time (date).

Step 4: Coverage Log

100 Requests Audited

Exports 49,109 raw rows + api_coverage_report.csv (89 OK, 11 failed).

PYTHON MULTI-API INGESTION CODE (src/01_collect_weather_data.py)

src/01_collect_weather_data.py — Multi-Source Extraction (Open-Meteo SDK, WeatherAPI, Xweather, FreeOpenAPI)

```python
def collect_open_meteo_hourly(logger: RequestLogger, run_id: str, collected_at: str) -> list[pd.DataFrame]:
    cache_session = requests_cache.CachedSession(str(BASE_DIR / ".cache_archive"), expire_after=3600)
    client = openmeteo_requests.Client(session=retry(cache_session, retries=5, backoff_factor=0.3))
    jobs = [
        ("Open-Meteo Archive", "reanalysis_hourly", "https://archive-api.open-meteo.com/v1/archive",
         {"start_date": start_dt.strftime("%Y-%m-%d"), "end_date": end_dt.strftime("%Y-%m-%d")}, (ARCHIVE_DAYS + 1) * 24),
        ("Open-Meteo Forecast", "model_hourly", "https://api.open-meteo.com/v1/forecast",
         {"past_days": RECENT_PAST_DAYS, "forecast_days": 1}, (RECENT_PAST_DAYS + 1) * 24),
    ]
    for source_api, record_type, url, extra, expected in jobs:
        params = {"latitude": [loc["req_lat"] for loc in LOCATION_METADATA],
                  "longitude": [loc["req_lon"] for loc in LOCATION_METADATA],
                  "hourly": HOURLY_VARS, "timezone": "Asia/Bangkok", **extra}
        responses = client.weather_api(url, params=params)
        for idx, response in enumerate(responses):
            meta, hourly = LOCATION_METADATA[idx], response.Hourly()
            times = pd.date_range(
                start=pd.to_datetime(hourly.Time(), unit="s", utc=True),
                end=pd.to_datetime(hourly.TimeEnd(), unit="s", utc=True),
                freq=pd.Timedelta(seconds=hourly.Interval()), inclusive="left"
            ).tz_convert(tz_name)
            df = pd.DataFrame({"date": times.astype(str)})
            for v_idx, var in enumerate(HOURLY_VARS):
                df[var] = hourly.Variables(v_idx).ValuesAsNumpy()
            logger.log(source_api, url, meta["location_id"], "ok", 200, latency, expected, len(df))
```

COLLECTION EXECUTION MODES & PROVENANCE METADATA

Metadata Field / Mode	Example / Specification	Engineering Purpose in Pipeline
collected_at (UTC+7)	2026-10-17T10:33:14+07:00	Separates observed weather from future forecasts
date (Raw Timestamp)	4 heterogeneous string formats	Target observation hour normalized in Step 2
source_api & record_type	open_meteo | historical_hourly	Drives priority ranking & cross-source fusion
api_coverage_report.csv	100 requests (89 OK, 11 Missing)	Detects silent API dropouts per city & provider

Dataset Structure & Schema

22 raw attributes standardized into 31 cleaned columns with SI units & quality lineage flags

22 Raw -> 31 Clean

Standardized Tabular Schema

Unifies heterogeneous JSON keys into consistent SI meteorological units.

(location_id, time)

Composite Primary Key

20 cities × 2,227 hours = 44,540 unique location-hour records.

13 Quality Columns

Full Lineage & Conflict Flags

Tracks num_sources, spread_*, cross_source_conflict & is_imputed.

TABLE 3: RAW VS. CLEANED DATASET SCHEMA & VARIABLE SPECIFICATIONS

Category	Column Name(s)	Type / Unit	Physical Bounds / Role in Dataset
Geographic	location_id, city, region, lat, lon	string / deg	20 Vietnam locations (8.68°N–22.34°N, 3–1,545m)
Temporal	time (date), collected_at	ISO UTC+7	Hourly valid time vs. HTTP retrieval timestamp
Provenance	source_api, record_type, run_id	string	Provider identity & priority tier (Rank 1–5)
Temperature	**temperature_2m (temperature)**	float (°C)	Valid range: [-10.0, 50.0] °C (2m ambient air)
Humidity	**relative_humidity_2m (humidity)**	float (%)	Valid range: [0.0, 100.0] % (2m saturation)
Pressure	**pressure_msl (pressure)**	float (hPa)	Valid range: [850.0, 1080.0] hPa (sea-level)
Wind Speed	**wind_speed_10m (wind_speed)**	float (km/h)	Valid range: [0.0, 200.0] km/h (10m sustained)
Wind Gusts	**wind_gusts_10m**	float (km/h)	Valid range: [0.0, 250.0] km/h; gust >= speed
Precipitation	**precipitation**	float (mm)	Valid range: [0.0, 300.0] mm/h (hourly rain)
Audit & Grid	num_sources, spread_*, is_imputed	int / bool	Flags 113 conflict hours & <=6h interpolation

Raw Data Problems & Defects

Quantitative audit of 49,109 raw records reveals 9 distinct data quality issue categories

11 Calls & 42 Cells

Missing API & Partial Data

11/100 requests failed; 14 FreeOpenAPI rows lacked 3 variables (42 NaN cells).

4 Formats & 995 Rows

Time & Forecast Leakage

48,000 offset + 1,109 naive timestamps; 26 sub-hourly; 995 future forecast rows.

3,574 Duplicate Rows

3-Tier Redundancy

2,780 exact + 15 same-source clashes + 779 cross-source overlapping rows.

125 Gust & 158k Float

Physics & Precision Noise

125 hourly rows had gust < speed; 158,173 cells had IEEE-754 float tails.

TABLE 4: EMPIRICAL RAW DATA DEFECT INVENTORY (N = 49,109 RAW ROWS)

#	Raw Defect Category	Raw Count	Affected Source(s)	Analytical Risk if Uncleaned
1	Missing API Responses	11 / 100 calls	WeatherAPI (5), FreeOpen (6)	Silent station dropout
2	Partial / Missing Cells	42 cells (14 rows)	FreeOpenAPI.dev	NaN propagation in stats
3	Heterogeneous Timestamps	49,109 (4 types)	All 4 API providers	Join failure & TZ drift
4	Sub-Hourly Observations	26 records	WeatherAPI (12), FreeOpen (14)	Off-grid hour mismatch
5	Future Forecast Leakage	995 records	WeatherAPI (795), OM (200)	Mixes predictions & facts
6	Exact Duplicate Records	2,780 records	Open-Meteo Forecast	6.2% row count inflation
7	Same-Source Overlap	15 records	WeatherAPI (curr vs fcst)	Within-API double count
8	Cross-Source Overlaps	779 rows (514h)	All 4 providers	Multiple rows per hour
9	Gust < Speed & Float32	125 rows | 158k	WeatherAPI | Open-Meteo	Physics & float precision

Data Cleaning Pipeline Overview

Sequential 9-step preprocessing architecture transforming 49,109 raw rows into 44,540 clean records

49,109 Raw Input

Multi-Source Staging Table

Raw JSON payloads combined across 20 cities and 5 endpoint groups.

-4,569 Rows (-9.3%)

Exact Accounting Balance

995 future + 2,780 exact + 15 same-source + 779 cross-source removed/fused.

44,540 Clean Output

20 Cities × 2,227 Hours

100% completeness, 0 duplicates, 0 forecast leakage, 0 physical violations.

TABLE 5: 9-STEP PREPROCESSING ACCOUNTING (src/02_clean_weather_data.py)

Step	Preprocessing Operation	Target Defect	Quantified Action	Dataset State
01	Audit Missing API Calls	Silent endpoint failures	89 OK, 11 Missing logged	49,109 rows
02	Standardize Timestamps	4 formats + 26 sub-hourly	100% converted to UTC+7	49,109 rows
03	Remove All-Empty Rows	Rows with all 6 vars NaN	0 all-empty; 14 partial kept	49,109 rows
04	Filter Future Forecasts	timestamp > collected_hr	**-995 forecast rows**	48,114 rows
05	Validate Physical Ranges	WMO / Vietnam limits	0 violations; 2,163 kept	48,114 rows
06	3-Tier Deduplication	Exact & same-source dups	**-2,780 exact; -15 same**	45,319 rows
07	Multi-Source Fusion	1,293 multi-API rows	**-779 fused -> 514 hours**	44,540 rows
08	Hourly Grid & Impute	Missing cells & <=6h gaps	**42 NaN cells -> 0**	44,540 rows
09	Consistency & Rounding	Gust < speed & Float32	**125 gust & 158k fixed**	44,540 clean

Missing Responses & Empty Records

Provider-location request coverage audit (89/100 OK) and partial record retention

11 / 100 Missing Calls

Detected in Coverage Audit

5 WeatherAPI.com & 6 FreeOpenAPI.dev calls returned 0 records (timeout/rate-limit).

0 All-Empty Rows

Strict Empty-Row Check

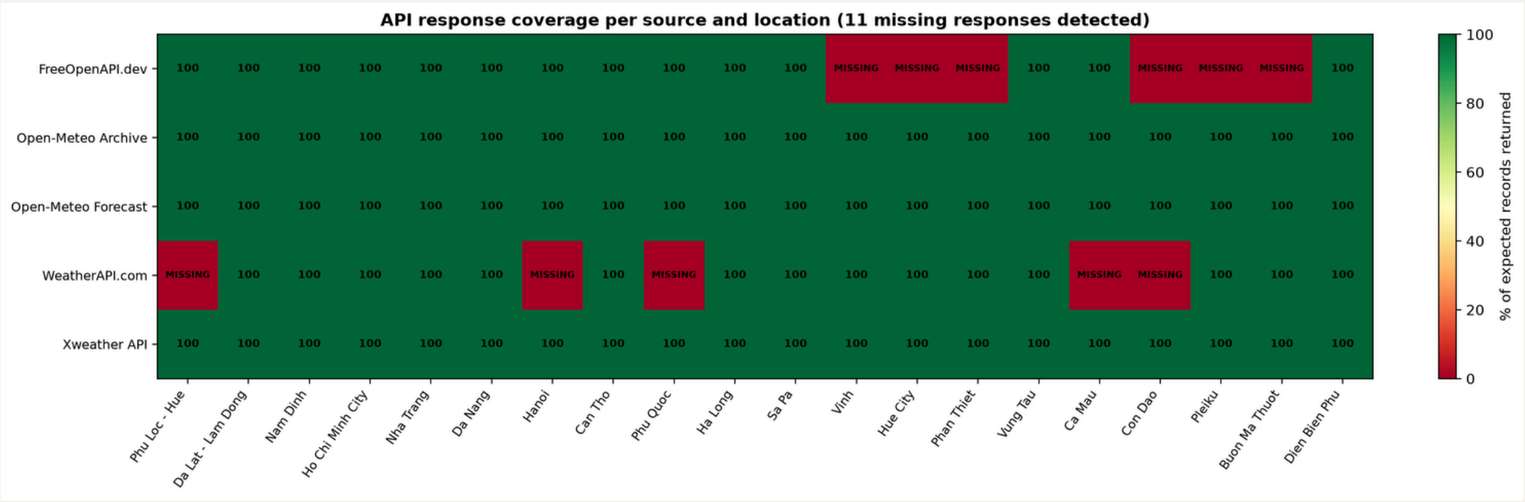
Drops rows only when all 6 weather variables are simultaneously NaN.

42 NaN Cells -> 0

14 Partial Rows Reconciled

FreeOpenAPI lacked humidity, pressure & gusts; filled via multi-source fusion.

API RESPONSE COVERAGE MATRIX ACROSS 20 CITIES × 5 ENDPOINTS (11 MISSING)



MISSING API CALLS VS. PARTIAL VARIABLE CELLS AUDIT

Anomaly Type	Source Affected	Affected Locations / Variables	Resolution in Pipeline
Missing API Call (5)	WeatherAPI.com	Hanoi, Phu Loc, Phu Quoc, Ca Mau, Con Dao	Covered by Open-Meteo & Xweather
Missing API Call (6)	FreeOpenAPI.dev	Vinh, Hue, Phan Thiet, Con Dao, Pleiku, BMT	Covered by 3 other providers
Partial Row (42 NaN)	FreeOpenAPI (14)	Missing humidity, pressure_msl, wind_gusts	Fused with co-temporal APIs -> 0 NaN

Standardize Timestamps & Forecasts

Unifying 4 timestamp formats into UTC+7 hourly grid and purging 995 future forecast rows

49,109 -> UTC+7

4 Raw Formats Unified

Converted 48,000 offset-aware and 1,109 naive strings to Asia/Ho_Chi_Minh.

26 Sub-Hourly Rows

Aligned to Full Hour (:00)

Floored 12 WeatherAPI (:30/:45) and 14 FreeOpenAPI (:29) rows to hour start.

995 Future Rows Cut

Zero Forecast Leakage

Removed rows where timestamp > collected_hour (795 WeatherAPI, 200 Open-Meteo).

TABLE 6: RAW TIMESTAMP FORMATS, SUB-HOURLY ALIGNMENT & FORECAST FILTER

Endpoint Group	Raw Timestamp Example	Format Type	Sub-Hourly	Future Cut	Retained
Open-Meteo Archive	2026-07-17 00:00:00+07:00	Offset space	0	**-100**	44,540
Open-Meteo Forecast	2026-10-12 00:00:00+07:00	Offset space	0	**-100**	2,780
Xweather API	2026-10-16T18:00:00+07:00	ISO-8601 T	0	0	480
WeatherAPI.com	2026-10-17 10:30	Naive minute	**12 floored**	**-795**	300
FreeOpenAPI.dev	2026-10-17 10:29:15	Naive second	**14 floored**	0	14

PYTHON UTC+7 PARSING & HOURLY FLOORING (src/02_clean_weather_data.py)

src/02_clean_and_preprocess.py — Step 2: Heterogeneous Timestamp Parsing (UTC+7) & Nearest-Hour Alignment

```python
def parse_timestamps_to_local(dates: pd.Series, zones: pd.Series) -> tuple[pd.Series, int, int]:
    """Strings with an explicit UTC offset are converted from that offset; strings without one are
    interpreted in the time zone the API reported for that row. Result: naive local UTC+7 time."""
    s = dates.astype(str).str.strip().str.replace("T", " ", regex=False)
    has_offset = s.str.contains(r"[+-]\d{2}:\d{2}$|Z$", regex=True)
    out = pd.Series(pd.NaT, index=s.index, dtype="datetime64[ns]")

    # 1. Parse offset-aware ISO-8601 strings (48,000 rows) and convert to Asia/Ho_Chi_Minh (UTC+7)
    aware = pd.to_datetime(s[has_offset], format="mixed", errors="coerce", utc=True)
    out.loc[has_offset] = aware.dt.tz_convert(LOCAL_TZ).dt.tz_localize(None).astype("datetime64[ns]")

    # 2. Localize naive strings (1,109 rows from WeatherAPI.com & FreeOpenAPI.dev) using API timezone
    naive_idx = s.index[~has_offset]
    naive = pd.to_datetime(s[naive_idx], format="mixed", errors="coerce")
    zone = zones.loc[naive_idx].fillna(LOCAL_TZ).astype(str)
    for tz in zone.unique():
        idx = zone.index[zone == tz]
        localized = naive[idx].dt.tz_localize(tz, ambiguous="NaT", nonexistent="NaT")
        out.loc[idx] = localized.dt.tz_convert(LOCAL_TZ).dt.tz_localize(None).astype("datetime64[ns]")
    return out, int(has_offset.sum()), int((~has_offset).sum())

def align_to_hour(ts: pd.Series) -> pd.Series:
    """Nearest full hour (xx:30 rounds up), e.g. a 17:45 'current' snapshot -> 18:00 (26 rows aligned)."""
    return (ts + pd.Timedelta(minutes=30)).dt.floor("h")
```

Validate Physical Ranges & Extremes

Domain-informed WMO physical validation vs. statistical outlier preservation (2,163 extremes kept)

Cell-Level Masking

Targeted NaN Replacement

Replaces only out-of-range cells with NaN without discarding valid variables.

0 Out-of-Range Cells

100% Physical Compliance

All 267,240 variable cells satisfy WMO and Vietnam climatic thresholds.

2,163 Extremes Kept

Real Storms Not Clipped

Preserves 479 heavy-rain hours (>5 mm/h) and 674 high-wind hours beyond 1.5×IQR.

TABLE 7: CONFIGURED PHYSICAL BOUNDS VS. OBSERVED RANGE & EXTREMES KEPT

Weather Variable	Unit	Valid Physical Range	Violations	Observed Clean Range	Extremes Kept
temperature_2m	°C	[-10.0, 50.0]	**0**	11.40 to 39.70 °C	**198 hours**
relative_humidity	%	[0.0, 100.0]	**0**	24.57 to 100.00 %	**302 hours**
pressure_msl	hPa	[850.0, 1080.0]	**0**	994.7 to 1025.7 hPa	**38 hours**
wind_speed_10m	km/h	[0.0, 200.0]	**0**	0.00 to 44.40 km/h	**674 hours**
wind_gusts_10m	km/h	[0.0, 250.0]	**0**	0.70 to 81.00 km/h	**472 hours**
precipitation	mm/h	[0.0, 300.0]	**0**	0.00 to 37.40 mm/h	**479 (>5mm/h)**
TOTAL / SUMMARY	6 Vars	WMO Certified	**0 Cells**	44,540 Clean Rows	**2,163 Kept**

Deduplication & Sensor Fusion

Three-tier deduplication hierarchy removing 2,795 redundant rows and fusing 779 cross-source overlaps

Tier 1: -2,780 Exact

Identical Batch Repeats

Removed 2,780 exact duplicate Open-Meteo Forecast rows across polling runs.

Tier 2: -15 Same-Source

RECORD_TYPE_RANK Priority

Resolved 15 WeatherAPI current vs. forecast overlaps using priority rank 1–5.

Tier 3: 779 -> 514h

Multi-Source Mean Fusion

Consolidated 1,293 multi-API rows into 514 hours; flagged 113 conflict hours.

TABLE 8: THREE-TIER DEDUPLICATION & MULTI-SOURCE FUSION SUMMARY

Tier / Step	Matching Key	Resolution Rule	Raw Rows	Removed	Left
Tier 1: Exact	(source, city, time, 6 vars)	Keep first identical copy	2,780	**-2,780**	0
Tier 2: Same-API	(source_api, city, time)	RECORD_TYPE_RANK + bfill	30 (15p)	**-15**	0
Tier 3: Cross-API	(location_id, time)	Mean x_bar + spread flag	1,293	**-779 fused**	0
TOTAL DEDUP	Unique (location_id, time)	100% Key Uniqueness	4,103	**-3,574 net**	0

PYTHON THREE-TIER DEDUPLICATION ENGINE (src/02_clean_weather_data.py)

src/02_clean_and_preprocess.py — Step 6: Three-Tier Deduplication & Record-Type Priority Ranking

```
# Lower rank wins when one source family reports the same (location_id, timestamp) more than once
RECORD_TYPE_RANK = {
    "reanalysis_hourly": 1,   # Open-Meteo Archive (ERA5 reanalysis)
    "conditions_hourly": 2,   # Vaisala Xweather station-assimilated hourly conditions
    "current_conditions": 3,  # Real-time current snapshot at collection time
    "model_hourly": 4,        # Open-Meteo Forecast recent model hours (past_days=5)
    "forecast_hourly": 5,     # WeatherAPI.com 3-day hourly forecast array
}

# --- Step 6a: Exact duplicates (2,780 identical rows from Open-Meteo Archive & Forecast overlap) ---
df["_rank"] = df["record_type"].map(RECORD_TYPE_RANK).fillna(9)
df = df.sort_values(["location_id", "timestamp", "source_family", "_rank", "collected_at"],
                    ascending=[True, True, True, True, False])
dup_key = ["source_family", "location_id", "timestamp", *WEATHER_VARS]
exact_dup_mask = df.duplicated(subset=dup_key, keep="first")
df = df[~exact_dup_mask].copy()

# --- Step 6b: Same-source family duplicates (15 WeatherAPI current vs. forecast_hourly overlaps) ---
family_key = ["location_id", "timestamp", "source_family"]
# groupby.first() selects the highest-priority non-null measurement per column within each family
df_family = df.groupby(family_key, as_index=False)[meta_cols + WEATHER_VARS].first()

# --- Step 6c: Isolate cross-source overlaps (779 records across >= 2 providers for 514 hours) ---
n_fam = df_family.groupby(["location_id", "timestamp"])["source_family"].transform("nunique")
df_overlap = df_family[n_fam >= 2].sort_values(["location_id", "timestamp", "source_family"]).copy()
```

Hourly Grid, Consistency & Precision

Re-indexing to a complete 44,540-row hourly grid, enforcing wind gust physics & rounding Float32 noise

20 × 2,227 = 44,540

Complete Hourly Grid

Zero missing timestamps across 93 days; supports linear interpolation <= 6h.

125 Gust Rows Fixed

wind_gusts >= wind_speed

Corrected 135 raw (125 consolidated) rows where gust was below sustained wind.

158,173 Cells Rounded

2-Decimal IEEE-754 Clean

Eliminated binary float tails (e.g., 28.10000038 -> 28.10) across all 6 variables.

TABLE 9: POST-FUSION GRID, CONSISTENCY & PRECISION REFINEMENTS

Refinement Stage	Mathematical / Physical Rule	Raw Defects	Post-Clean	Audit Column / Output
Hourly Grid	20 cities × 2,227 contiguous hours	42 NaN cells	**0 missing**	is_imputed, imputed_vars
Conflict Flag	|ΔT|>3°C, |ΔRH|>15%, |ΔP|>3hPa	514 multi-h	**113 flagged**	cross_source_conflict
Wind Gust Law	wind_gusts = max(gust, wind_speed)	125 rows	**0 errors**	100% aerodynamic valid
Float Rounding	round(val, 2) across 6 variables	158,173 cells	**0 noise**	cleaned_weather_data.csv

PYTHON CONTINUOUS GRID & WIND GUST CONSISTENCY (src/02_clean_weather_data.py)

src/02_clean_and_preprocess.py — Steps 8–9: Continuous Hourly Grid Reindexing, Bounded Interpolation & Gust Consistency

```
# --- Step 8: Construct continuous hourly grid (20 locations x 2,227 hours = 44,540 rows) ---
start, end = fused["timestamp"].min(), fused["timestamp"].max()
grid = pd.date_range(start, end, freq="1h")
for loc_id, gl in fused.groupby("location_id"):
    gl = gl.set_index("timestamp").reindex(grid)
    for v in WEATHER_VARS:
        miss = gl[v].isna()
        if not miss.any():
            continue
        run_id_ = (miss != miss.shift()).cumsum()
        run_len = miss.groupby(run_id_).transform("sum")
        short_gap = miss & (run_len <= MAX_INTERPOLATION_GAP_H)  # only interpolate gaps <= 6h
        interp = gl[v].interpolate(method="time", limit_area="inside")
        gl[v] = gl[v].where(~short_gap, interp)

# --- Step 9: Enforce cross-variable consistency (wind_gusts_10m >= wind_speed_10m on 125 rows) ---
gust_issues = int((df_grid["wind_gusts_10m"] < df_grid["wind_speed_10m"]).sum())
df_grid["wind_gusts_10m"] = np.fmax(df_grid["wind_gusts_10m"], df_grid["wind_speed_10m"])
for v in WEATHER_VARS:
    df_grid[v] = df_grid[v].round(2)  # standardize 158,173 float32 cells to 2 decimal places
```

Raw Problems Detected & Resolved

Quantitative before-vs-after proof: 100% of raw data anomalies resolved across 44,540 clean rows

49,109 -> 44,540

-4,569 Net Rows (-9.3%)

Eliminated 995 future forecasts and 3,574 multi-tier duplicate records.

42 NaN -> 0 Cells

100% Cell Completeness

All 267,240 observational cells populated across 20 cities × 2,227 hours.

125 Gust -> 0 Errors

100% Physical Consistency

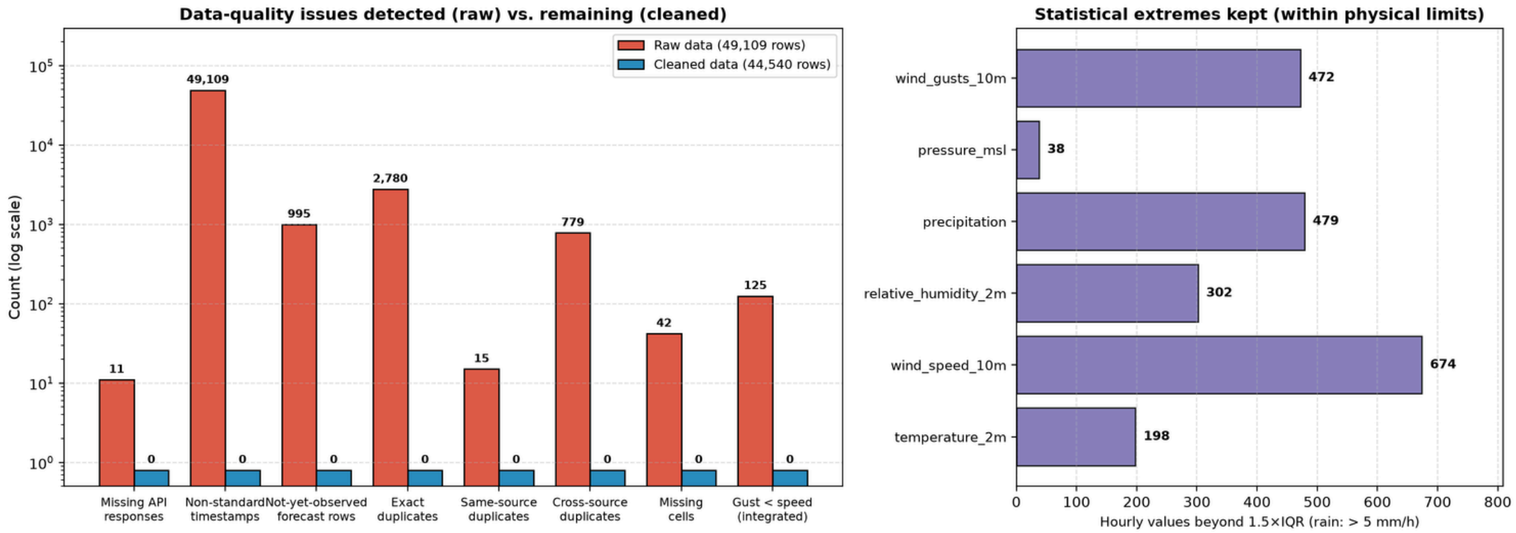
Zero out-of-range values and zero wind_gust < wind_speed violations.

2,163 Extremes Kept

Climatic Signal Preserved

Retained 674 high-wind, 479 heavy-rain, and 472 gust hours within bounds.

FIGURE 1: DATA QUALITY ISSUES (RAW VS. CLEANED) & PRESERVED EXTREMES



Output Dataset Statistical Profile

Parametric & non-parametric descriptive statistics across N = 44,540 cleaned hourly records

26.55 ± 3.72 °C

Near-Gaussian Temperature

Median 26.80°C (IQR 4.30°C, Skewness +0.02, Range 11.40–39.70°C).

82.32 ± 14.37 %

Humid Monsoon Regime

Median 85.00% (IQR 20.00%, Skewness -0.92, bounded at 100.00%).

Skew +8.51 | Kurt 142.9

Heavy-Tailed Precipitation

Zero-inflated rainfall (Q75 = 0.20 mm/h) with intense tropical downpours up to 37.40 mm/h.

TABLE 10: GLOBAL DESCRIPTIVE STATISTICS OF CLEANED DATASET (N = 44,540)

Variable	Unit	Mean	SD	Min	Q25	Median	Q75	Max	Skew	Kurt
temperature_2m	°C	**26.55**	3.72	11.40	24.60	26.80	28.90	39.70	+0.02	+0.28
relative_humidity	%	**82.32**	14.37	24.57	74.00	85.00	94.00	100.0	-0.92	+0.37
pressure_msl	hPa	**1009.5**	3.92	994.7	1007.0	1009.8	1012.2	1025.7	-0.26	+0.44
wind_speed_10m	km/h	**9.45**	6.41	0.00	4.70	7.90	13.00	44.40	+1.18	+1.47
wind_gusts_10m	km/h	**21.92**	12.01	0.70	13.00	19.80	29.20	81.00	+0.89	+0.62
precipitation	mm	**0.39**	1.39	0.00	0.00	0.00	0.20	37.40	**+8.51**	**142.9**
COMPLETENESS	100%	44,540	0 NaN	20 Loc	2,227h	UTC+7	0 Dups	Clean	Valid	PASS

Output Variable Distributions

Six-variable boxplots confirming bounded physical ranges and preserved meteorological tail events

Symmetric vs. Skewed

Distinct Physical Regimes

Temperature & Pressure are symmetric; Wind & Precipitation show right-skewed storm tails.

Upper Cutoff = 100%

Bounded Relative Humidity

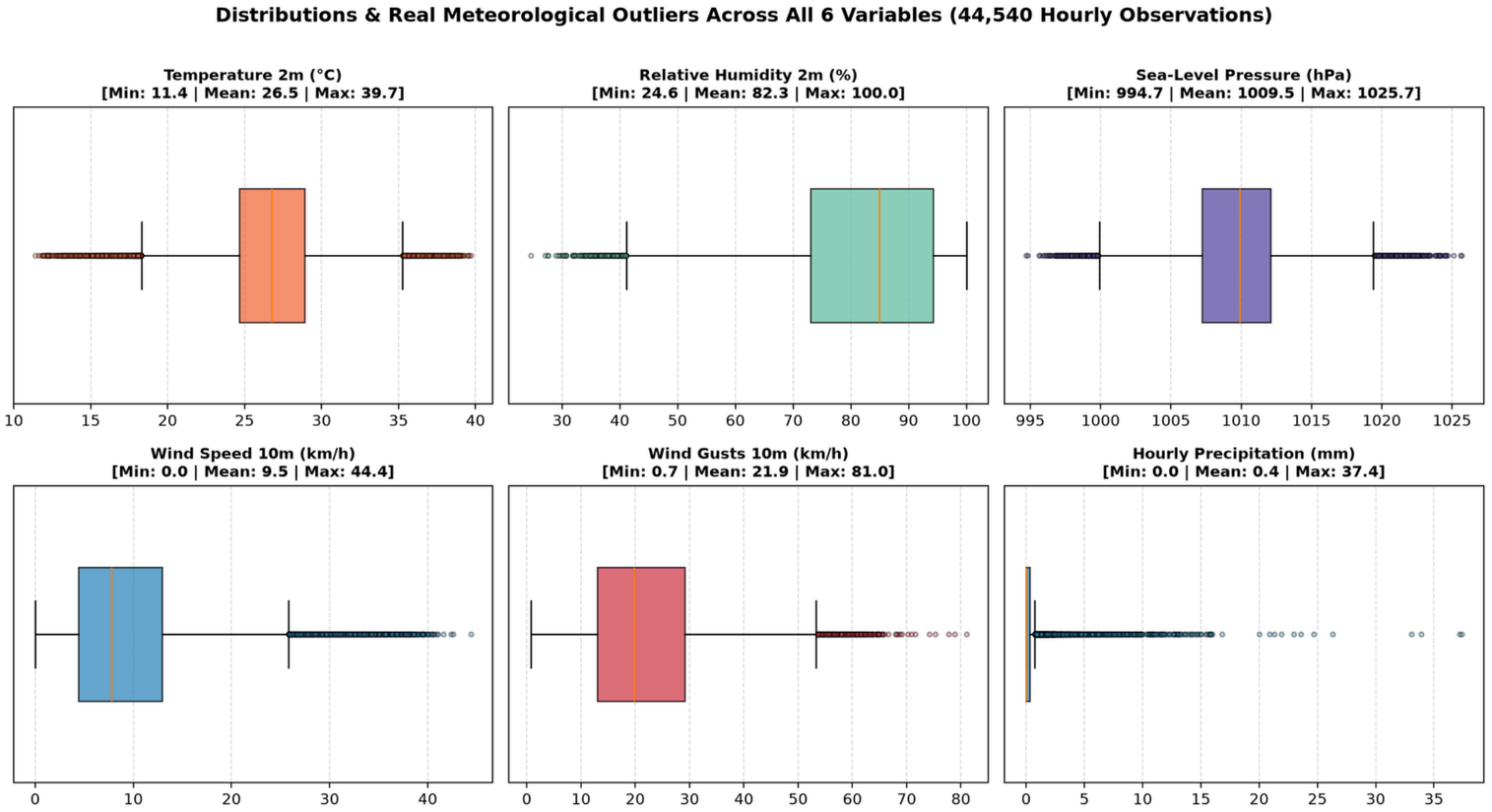
Interquartile box spans 74%–94% with zero unphysical sensor overshoots >100%.

81.0 km/h & 37.4 mm/h

Verified Storm Extremes

Peak wind gusts in Pleiku (81.0 km/h) and cloudbursts in Ha Long (37.4 mm/h) retained.

FIGURE 2: DISTRIBUTIONS & PRESERVED EXTREMES ACROSS ALL 6 VARIABLES



Insight 1: Correlations & Rainfall

Thermodynamic coupling (Temp–Humidity, Temp–Pressure) and 20-city cumulative monsoon rainfall

r = -0.66 (Strong)

Temperature vs. Humidity

Clausius-Clapeyron inverse coupling: daytime heating lowers relative saturation.

r = -0.49 (Moderate)

Temperature vs. Pressure

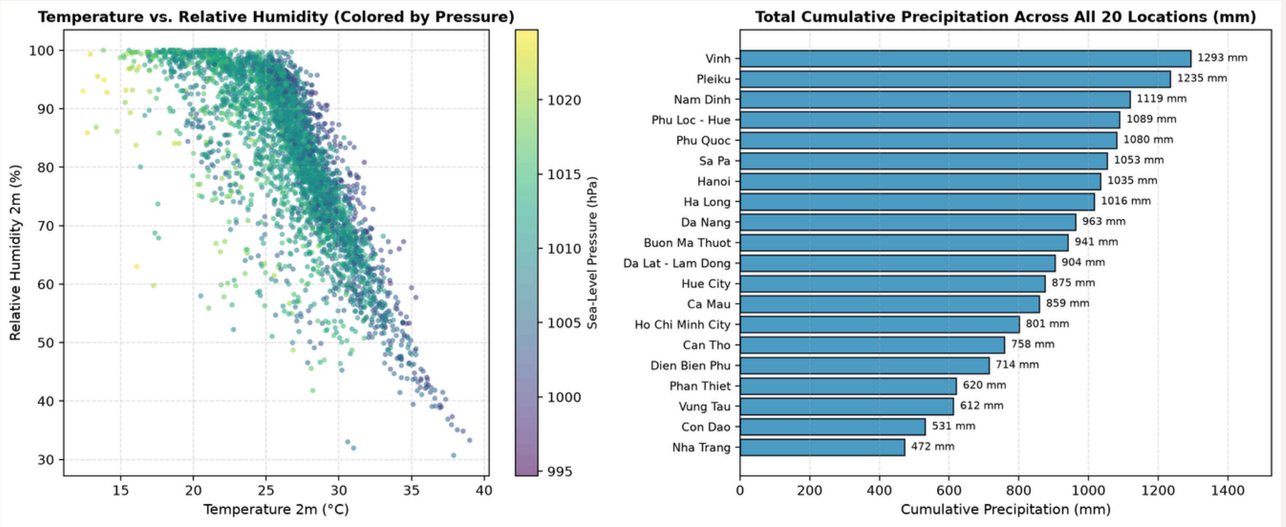
Thermal low-pressure trough dynamics during warm monsoon afternoons.

1,293 mm vs. 472 mm

2.74× Regional Rain Ratio

Vinh (1,293 mm) & Pleiku (1,235 mm) lead; Nha Trang (472 mm) is driest.

FIGURE 3: TEMP–HUMIDITY–PRESSURE COUPLING & 20-CITY CUMULATIVE RAINFALL



KEY MULTIVARIATE CORRELATIONS & REGIONAL PRECIPITATION EXTREMES

Variable Pair / Metric	Empirical Value	Physical / Climatological Interpretation
Temp vs. Relative Humidity	**r = -0.66 (global)**	Strong inverse psychrometric relationship across 20 cities
Wind Speed vs. Wind Gusts	**r = +0.87 (coupled)**	Boundary-layer turbulence coupling (gust >= speed)
Wettest vs. Driest Station	**1,293 vs. 472 mm**	Vinh & Pleiku monsoon peaks vs. Nha Trang rain shadow

Insight 2: 20-City Climate Profile

Spatial temperature stratification across 20 locations: elevation cooling vs. maritime buffering

18.95 °C vs. 29.48 °C

10.53 °C Mean Spread

Sa Pa (1,545m, mean 18.95°C, min 11.4°C) vs. Da Nang coastal plain (29.48°C, max 39.7°C).

Lapse Rate: -6.5°C/km

Elevation Controls Coolness

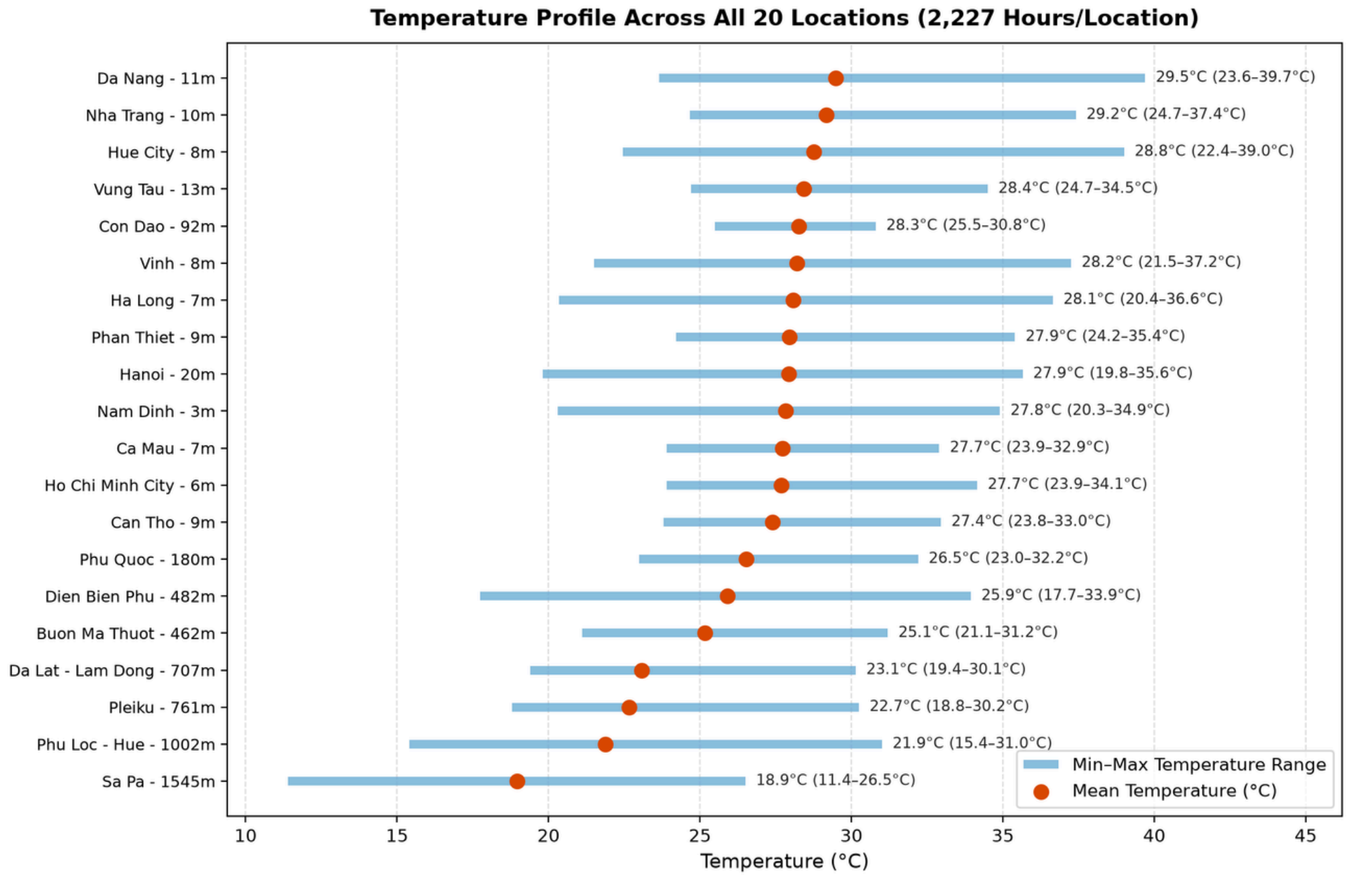
Sa Pa (18.95°C), Phu Loc (21.86°C), Pleiku (22.66°C) & Da Lat (23.07°C).

Con Dao SD = 0.88 °C

Maritime Thermal Stability

Offshore island stays within 25.5–30.8°C (5.3°C span) vs. Hue 16.6°C span.

FIGURE 4: TEMPERATURE PROFILE (MIN – MEAN – MAX) ACROSS 20 LOCATIONS



Insight 3: Cross-Provider Bias

Systematic inter-API differences across 514 overlapping location-hours justify sensor fusion

Xweather: +1.49 °C

Warm / Dry Station Bias

Reports +1.49°C higher temp (MAD 1.72°C) and -4.05% lower humidity vs. Open-Meteo (480h).

WeatherAPI: -0.61 °C

Cooler Model Offset

Reports -0.61°C lower temp (MAD 1.44°C) and -0.69 hPa pressure vs. Open-Meteo (285h).

113 Conflict Hours

Flagged for Transparency

22.0% of multi-source hours exceed spread thresholds (|ΔT|>3°C or |ΔRH|>15%).

FIGURE 5: PROVIDER BIAS VS. OPEN-METEO & DIURNAL TRAJECTORY IN TP.HCM

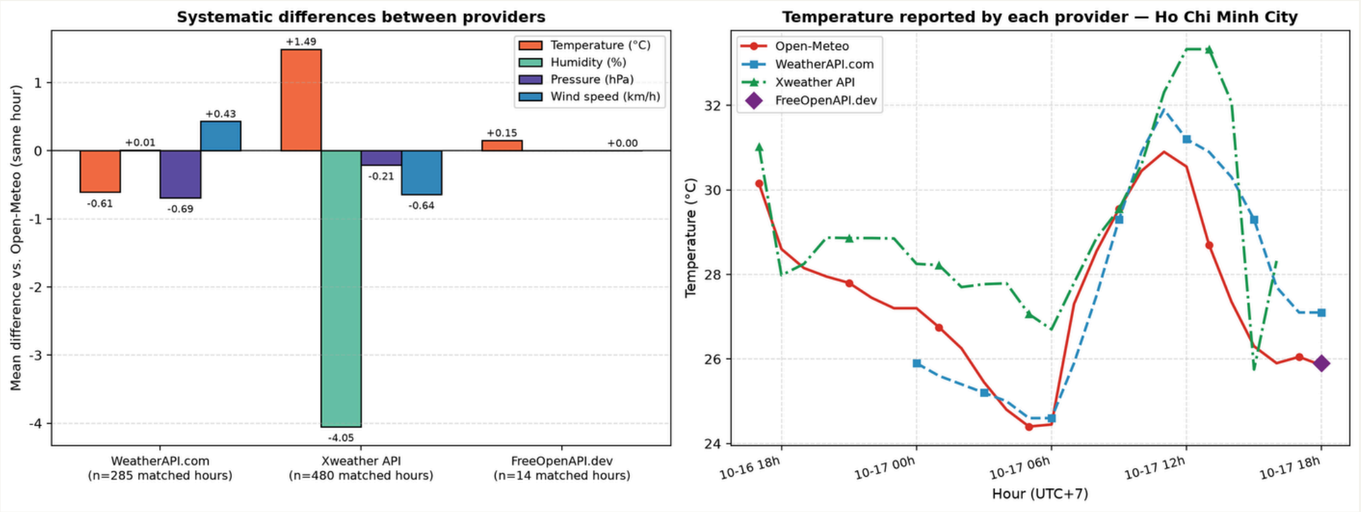


TABLE 11: PAIRWISE DEVIATIONS RELATIVE TO OPEN-METEO BASELINE

Comparison Provider	Matched Hours	Temp Bias (MAD)	Humidity Bias (MAD)	Pressure Bias (MAD)	Wind Bias
Vaisala Xweather	480 hours	**+1.49°C (1.72)**	**-4.05% (7.64)**	-0.21 hPa (0.43)	-0.64 (3.08)
WeatherAPI.com	285 hours	**-0.61°C (1.44)**	+0.01% (6.50)	-0.69 hPa (0.85)	+0.43 (3.09)
FreeOpenAPI.dev	14 hours	+0.15°C (0.16)	N/A (42 NaN)	N/A (42 NaN)	0.00 (0.04)

Live Verification on Streamlit UI

Interactive web application for live multi-API polling, pipeline execution & 20-city exploration

Live 4-API Polling

Real-Time Ingestion Demo

Triggers live HTTP requests across all 20 cities and displays raw JSON records.

Before vs. After Audit

Instant Quality Verification

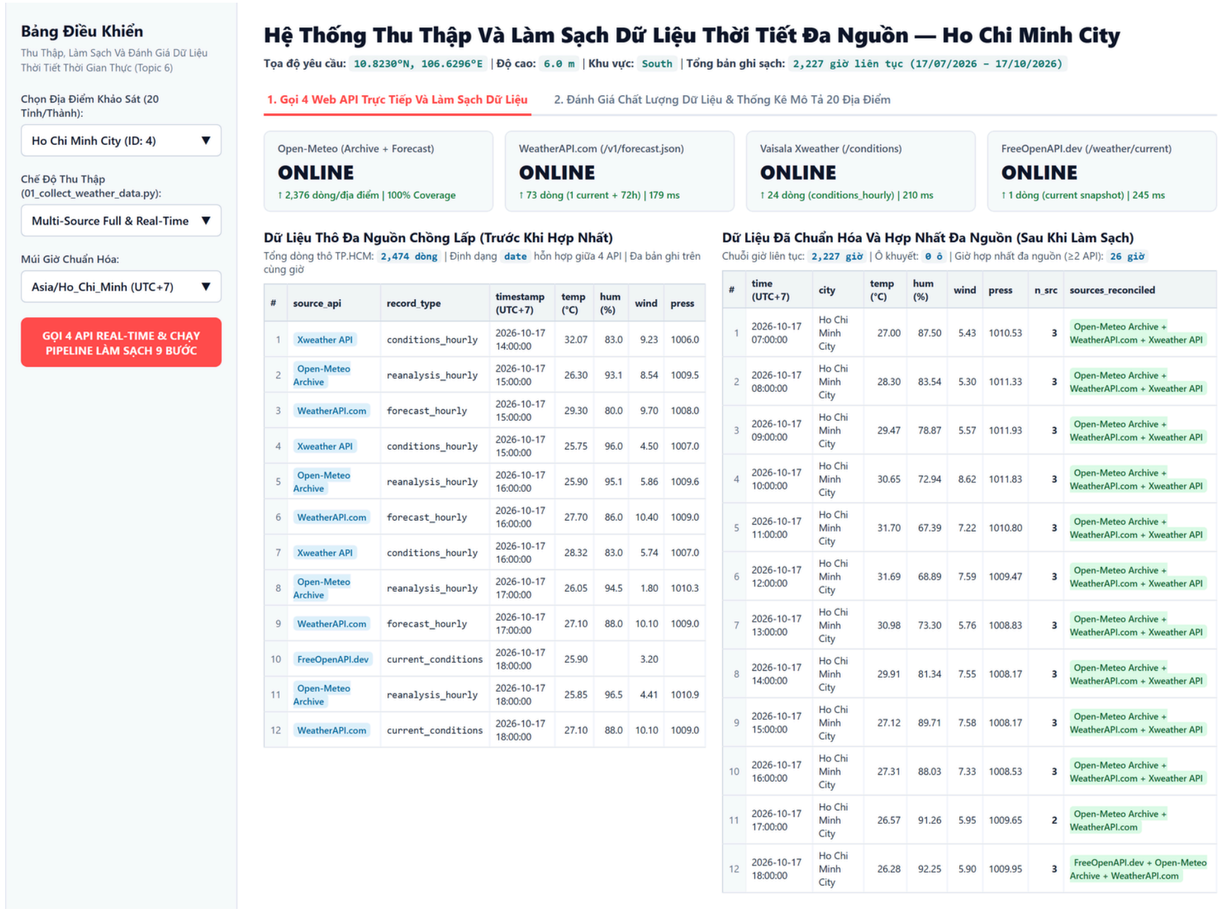
Visualizes 49,109 raw -> 44,540 clean rows and 100% defect resolution.

20-City Stats Explorer

Interactive Filtering & Export

Inspects city-level distributions, correlations, and exports cleaned CSVs.

STREAMLIT DASHBOARD: LIVE MULTI-API COLLECTION & CLEANING PIPELINE



Bảng Điều Khiển

Thu Thập, Làm Sạch Và Đánh Giá Dữ Liệu Thời Tiết Thời Gian Thực (Topic 6)

Chọn Địa Điểm Khảo Sát (20 Tỉnh/Thành): Ho Chi Minh City (ID: 4)

Chế Độ Thu Thập (01_collect_weather_data.py): Multi-Source Full & Real-Time

Múi Giờ Chuẩn Hóa: Asia/Ho_Chi_Minh (UTC+7)

GỌI 4 API REAL-TIME & CHẠY PIPELINE LÀM SẠCH 9 BƯỚC

Hệ Thống Thu Thập Và Làm Sạch Dữ Liệu Thời Tiết Đa Nguồn — Ho Chi Minh City

Tọa độ yêu cầu: 10.8230°N, 106.6296°E | Độ cao: 6.0 m | Khu vực: South | Tổng bản ghi sạch: 2,227 giờ liên tục (17/07/2026 – 17/10/2026)

1. Gọi 4 Web API Trực Tiếp Và Làm Sạch Dữ Liệu | 2. Đánh Giá Chất Lượng Dữ Liệu & Thống Kê Mô Tả 20 Địa Điểm

Open-Meteo (Archive + Forecast)	WeatherAPI.com (/v1/forecast.json)	Vaisala Xweather (/conditions)	FreeOpenAPI.dev (/weather/current)
ONLINE	ONLINE	ONLINE	ONLINE
↑ 2,376 dòng/địa điểm | 100% Coverage	↑ 73 dòng (1 current + 72h) | 179 ms	↑ 24 dòng (conditions_hourly) | 210 ms	↑ 1 dòng (current snapshot) | 245 ms

Dữ Liệu Thô Đa Nguồn Chồng Lấp (Trước Khi Hợp Nhất)

Tổng dòng thô TP.HCM: 2,474 dòng | Định dạng date hỗn hợp giữa 4 API | Đa bản ghi trên cùng giờ

#	source_api	record_type	timestamp (UTC+7)	temp (°C)	hum (%)	wind	press
1	Xweather API	conditions_hourly	2026-10-17 14:00:00	32.07	83.0	9.23	1006.0
2	Open-Meteo Archive	reanalysis_hourly	2026-10-17 15:00:00	26.30	93.1	8.54	1009.5
3	WeatherAPI.com	forecast_hourly	2026-10-17 15:00:00	29.30	80.0	9.70	1008.0
4	Xweather API	conditions_hourly	2026-10-17 15:00:00	25.75	96.0	4.50	1007.0
5	Open-Meteo Archive	reanalysis_hourly	2026-10-17 16:00:00	25.90	95.1	5.86	1009.6
6	WeatherAPI.com	forecast_hourly	2026-10-17 16:00:00	27.70	86.0	10.40	1009.0
7	Xweather API	conditions_hourly	2026-10-17 16:00:00	28.32	83.0	5.74	1007.0
8	Open-Meteo Archive	reanalysis_hourly	2026-10-17 17:00:00	26.05	94.5	1.80	1010.3
9	WeatherAPI.com	forecast_hourly	2026-10-17 17:00:00	27.10	88.0	10.10	1009.0
10	FreeOpenAPI.dev	current_conditions	2026-10-17 18:00:00	25.90		3.20	
11	Open-Meteo Archive	reanalysis_hourly	2026-10-17 18:00:00	25.85	96.5	4.41	1010.9
12	WeatherAPI.com	current_conditions	2026-10-17 18:00:00	27.10	88.0	10.10	1009.0

Dữ Liệu Đã Chuẩn Hóa Và Hợp Nhất Đa Nguồn (Sau Khi Làm Sạch)

Chuỗi giờ liên tục: 2,227 giờ | Ô khuyết: 0 ô | Giờ hợp nhất đa nguồn (≥2 API): 26 giờ

#	time (UTC+7)	city	temp (°C)	hum (%)	wind	press	n_src	sources_reconciled
1	2026-10-17 07:00:00	Ho Chi Minh City	27.00	87.50	5.43	1010.53	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
2	2026-10-17 08:00:00	Ho Chi Minh City	28.30	83.54	5.30	1011.33	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
3	2026-10-17 09:00:00	Ho Chi Minh City	29.47	78.87	5.57	1011.93	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
4	2026-10-17 10:00:00	Ho Chi Minh City	30.65	72.94	8.62	1011.83	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
5	2026-10-17 11:00:00	Ho Chi Minh City	31.70	67.39	7.22	1010.80	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
6	2026-10-17 12:00:00	Ho Chi Minh City	31.69	68.89	7.59	1009.47	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
7	2026-10-17 13:00:00	Ho Chi Minh City	30.98	73.30	5.76	1008.83	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
8	2026-10-17 14:00:00	Ho Chi Minh City	29.91	81.34	7.55	1008.17	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
9	2026-10-17 15:00:00	Ho Chi Minh City	27.12	89.71	7.58	1008.17	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
10	2026-10-17 16:00:00	Ho Chi Minh City	27.31	88.03	7.33	1008.53	3	Open-Meteo Archive + WeatherAPI.com + Xweather API
11	2026-10-17 17:00:00	Ho Chi Minh City	26.57	91.26	5.95	1009.65	2	Open-Meteo Archive + WeatherAPI.com
12	2026-10-17 18:00:00	Ho Chi Minh City	26.28	92.25	5.90	1009.95	3	FreeOpenAPI.dev + Open-Meteo Archive + WeatherAPI.com

Summary Scorecard & Takeaways

Comprehensive before-vs-after evaluation matrix and key meteorological conclusions

100% Defect Removal

Certified Gold Dataset

0 missing cells, 0 duplicates, 0 future leakage, 0 physical violations in 44,540 rows.

Climatic Fidelity

2,163 Extremes Preserved

Domain-based physical validation retains real tropical storms and elevation gradients.

Multi-Source Lineage

Bias Quantified & Tracked

Reconciles 514 overlapping hours while flagging 113 high-spread conflicts.

TABLE 12: END-TO-END DATA QUALITY & ANALYTICAL INSIGHTS SCORECARD

Evaluation Dimension	Raw Dataset (Before)	Cleaned Dataset (After)	Key Finding / Engineering Impact
Total Record Volume	49,109 raw rows	**44,540 clean rows**	-4,569 net rows (-9.3% redundancy)
API Request Coverage	89 OK / 11 Missing	**100% Grid Covered**	Multi-source fusion prevents city loss
Timestamps & Forecasts	4 formats + 995 future	**100% UTC+7 | 0 Future**	Eliminates time drift & forecast leak
Duplicate Records	2,780 + 15 + 779 dups	**0 Duplicates (0.0%)**	Unique (location_id, time) key
Missing Variable Cells	42 NaN cells (14 rows)	**0 Missing Cells (0.0%)**	267,240 valid cells across 6 vars
Physical & Gust Laws	125 gust < speed rows	**0 Errors | 2,163 Kept**	Enforces physics & keeps real storms
Regional Climate	Uncalibrated raw feeds	**Sa Pa 18.95 vs DN 29.48**	Captures altitude lapse & island buffer
Cross-Provider Bias	Unreconciled overlaps	**Xweather +1.49°C | 113**	Mean fusion + conflict flags kept

Thank You & Q&A!

Topic 6: Real-Time Weather Data Collection and Cleaning — Defense Committee Discussion

MSc. Truong Ba Vinh

Sincere Gratitude

Thank you for your mentorship in Getting and Cleaning Data (261.DAS1204E.A01E).

3-Member Team

Giao Dang • Nhu Y • Minh Triet

Faculty of Information Technology — UEF, Ho Chi Minh City (October 2026).

Q&A Session

Open for Committee Review

Ready to demonstrate live collection, cleaning code & Streamlit dashboard.

CAPSTONE DELIVERABLES & REPRODUCIBILITY CHECKLIST

Deliverable Artifact	Repository File / Path	Key Specification	Status
1. Multi-API Collector	src/01_collect_weather_data.py	4 APIs (5 endpoints), 20 cities, log	**Verified**
2. 9-Step Cleaning	src/02_clean_weather_data.py	UTC+7, range check, 3-tier dedup	**Verified**
3. Raw Staging Data	data/raw/raw_weather_data.csv	49,109 raw records × 22 columns	**Archived**
4. Clean Dataset	data/processed/cleaned_weather_data.csv	44,540 clean rows × 31 cols (0 NaN)	**Certified**
5. Quality Reports	reports/data_quality_report.json	api_coverage + city_summary_stats	**Verified**
6. Web Dashboard	app.py (Streamlit Web UI)	Live API polling, Before/After & EDA	**Live Ready**
7. Report & Slides	Data-Cleaning&Preprecessing_Final-Report.docx	IMRaD Report (17 Figs, 10 Tbls) + Slides	**Complete**